

MAI-Voice-2.1 Model Card

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Overview

MAI-Voice-2.1 is a prompted text-to-speech (TTS) model built in-house by the Microsoft AI team, generating high-fidelity, natural, and expressive speech across 23 languages and 26 locales. It captures human-like intonation, rhythm, and emotional nuance, delivers emotional flexibility with turn-level control over tone and delivery, and can render a single voice identity consistently across every supported language, making it ideal for audiobooks, content creation, voice-over, media, and other scenarios where fidelity and expressiveness matter most.

Voice can be configured using:

- Curated voice library (licensed voices designed to work straight out of the box)
- Voice Cloning via short audio clips (5-60 seconds), and the model matches the voice instantly

Key capabilities

- Natural and expressive voice synthesis with realistic intonation, rhythm, and emotional range.
- High-fidelity, high-clarity voice output.
- Multilingual support across 23 languages.
- One voice across all supported languages: a single voice identity is rendered consistently in every supported language.
- Fine-grained control of tone and delivery: shape emotion, tone, and delivery at the turn/sentence level, including different speaking styles via SSML.
- State-of-the-art instant Voice Cloning: provide a short audio clip (5-60 seconds) and the model clones the voice instantly, with no fine-tuning required. Prompt quality significantly impacts output, with best results from natural, conversational delivery and moderate energy levels. Access requires Microsoft approval and guardrails are in place to avoid misuse.
- Long-form content generation with improved consistency via chunking and context carryover.
- Multi-speaker support for dialogue and conversational scenarios.

Voice Cloning safeguards

To ensure Voice Cloning is compliant, we adhere to Microsoft's Responsible AI policies and apply the following steps to ensure voice cloning is conducted lawfully.

Steps to Access Personal Voice (Voice Cloning) with MAI-Voice-2.1:

- Apply for gated access: Submit a request via the Azure AI Custom Neural Voice and Custom Avatar Limited Access Review.

- Access personal voice APIs: Once approved, use the APIs available at [cognitive-services-speech-sdk/samples/custom-voice](#)
- Upload audio consent and prompt: Provide a recorded audio consent statement from the voice talent along with the required prompt to create a personal voice profile. For more information, read about [Azure Speech consent statement](#).
- Synthesize text using the created voice

Use cases

Key use cases

- **Audiobooks and Media Narration:** Produce full-length, expressive narration with a consistent voice.
- **Content Creation and Voice-over:** Turn scripts into polished, high-fidelity spoken audio.
- **Marketing and Advertising:** Deliver a consistent brand voice across campaigns and markets.
- **E-learning and Educational Content:** Build interactive learning content with expressive narration.
- **Virtual Assistants and Chatbots:** Power conversational agents across apps and devices with natural voices.
- **Accessibility Features:** Provide narration for visually impaired users and assistive voice technologies.
- **Public Announcements:** Deliver clear, engaging voice output for public information systems.

Pricing

\$22 per 1M characters

Technical specs

Input formats

Transcript text with an optional speaker prompt for Voice Cloning (with 5-60s of audio), plus optional emotion tags like 'joyful' or 'relieved' via SSML.

SSML examples

Basic SSML Example (Harper):

```
<?xml version='1.0' xmlns='http://www.w3.org/2001/10/synthesis' xmlns:mstts='http://www.w3.org/2001/mstts'
xml:lang='en-US'>
  <voice name='en-US-Harper:MAI-Voice-2.1'>
    Hello world, this is awesome!
  </voice>
</speak>
```



Basic SSML Example (Grant):

```
<speak version='1.0' xmlns='http://www.w3.org/2001/10/synthesis' xmlns:mstts='http://www.w3.org/2001/mstts'
xml:lang='en-US'>
  <voice name='en-US-Grant:MAI-Voice-2.1'>
    Hello world, this is awesome!
  </voice>
</speak>
```

Multilingual SSML Example (Harper):

```
<speak version='1.0' xmlns='http://www.w3.org/2001/10/synthesis' xmlns:mstts='http://www.w3.org/2001/mstts'
xml:lang='en-US'>
  <voice name='en-US-Harper:MAI-Voice-2.1'> How are you? </voice>
  <voice name='hu-HU-Harper:MAI-Voice-2.1'> Hogy vagy? </voice>
  <voice name='vi-VN-Harper:MAI-Voice-2.1'> Bạn khỏe không? </voice>
  <voice name='es-MX-Harper:MAI-Voice-2.1'> ¿Cómo estás? </voice>
</speak>
```

Multilingual SSML Example (Grant):

```
<speak version='1.0' xmlns='http://www.w3.org/2001/10/synthesis' xmlns:mstts='http://www.w3.org/2001/mstts'
xml:lang='en-US'>
  <voice name='en-US-Grant:MAI-Voice-2.1'>I am fine, thank you</voice>
  <voice name='hu-HU-Grant:MAI-Voice-2.1'>Jól vagyok, köszönöm.</voice>
  <voice name='vi-VN-Grant:MAI-Voice-2.1'>Tôi ổn, cảm ơn bạn.</voice>
  <voice name='es-MX-Grant:MAI-Voice-2.1'>Estoy bien gracias.</voice>
</speak>
```

Expressive Control with SSML `mstts:express-as`:

```
<speak version='1.0' xmlns='http://www.w3.org/2001/10/synthesis' xmlns:mstts='http://www.w3.org/2001/mstts'
xml:lang='en-US'>
  <voice name='en-US-Grant:MAI-Voice-2.1'>
    <mstts:express-as style="excited">
      Hello world, this is awesome!
    </mstts:express-as>
  </voice>
</speak>
```

Input formats

Transcript text with an optional speaker prompt for Voice Cloning (with 5-60s of audio), plus optional emotion tags like 'joyful' or 'relieved' via SSML.

Output

Users can specify the desired output format. More information [here](#).

Supported languages

23 languages:

English (US), English (Australia), English (UK), English (India), Italian, French, German, Hindi, Spanish (Spain), Spanish (Mexico), Portuguese (Brazil), Portuguese (Portugal), Korean, Chinese (Simplified), Turkish, Russian, Thai, Dutch, Romanian, Hungarian, Czech, Danish, Finnish, Indonesian, Polish, Swedish, Norwegian (Bokmål), Vietnamese.



Distribution

Ready to integrate? Explore the full [MAI-Voice documentation](#) to get started.

MAI-Voice-2.1 is available via the Azure Speech SDK and REST APIs:

- Speech SDK: Integrate TTS capabilities directly into applications using Azure's Speech SDK, available for platforms including .NET, Python, Java, JavaScript, and C++.
- REST API: Access TTS functionality via a public, subscription-based API for flexible integration into web services, mobile apps, and backend systems.

MAI-Voice-2.1 can be accessed globally. The model is currently served from the following Azure regions to which the requests are routed:

- France Central
- East Asia
- Southeast Asia
- East US
- Canada Central
- East US 2
- West US
- West Europe
- North Europe
- West US 2
- West US 3
- Central India
- Sweden Central
- Japan East